

ALI NIKKHAH

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SUMMARY

Applied ML researcher & engineer (Sharif B.Sc. Electrical Engineering 2020-2025, GPA 3.65; M.Sc. AI from Sep 2025) specialising in agentic RAG, multimodal vision-language, self-hosted LLM serving, and GPU runtime optimisation. Shipped LLM + CV pipelines at marketplace scale, self-hosted Graph RAG assistant for a national credit bureau, and BNPL risk platforms. Research focus: LLM reasoning and hallucination mitigation. — Variant: Data Science/Eng

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL, R, Bash, TypeScript
ML/Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper
Agentic: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE
LLM Serving/GPU: vLLM, TGI, AWQ, GPTQ, KV-cache
Data/MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray

PROFESSIONAL EXPERIENCE

Turquoise Digital

Tehran, Iran

Senior Data Scientist & Data Engineer (Concurrent)

Sep 2025 == Mar 2026

- Centralized lake from raw CDC via PeerDB + Trino/Parquet on 5TB/day, <60s freshness; Airflow + PySpark DAGs cut memory 60%, governed by Great Expectations + Prometheus/Grafana.
- Built default-risk ensemble (XGBoost + time-series Transformers) — AUC 0.87, -22% defaults; 50+ concurrent mSPRT A/B tests with canary & auto-rollback, presented to C-suite.
- Shipped Whisper/wav2vec2.0 + LangGraph agentic chatbot — -78% ticket volume, 94% intent accuracy; deterministic fallback + policy guardrails for auditability.
- Drove portfolio allocation & risk engine (Sharpe +0.42) with streaming ClickHouse signals; LlamaIndex retrieval graph for ranking features.
- *Stack:* Python, PyTorch, Spark, Trino, PeerDB, Airflow, Great Expectations, Prometheus
- *Concurrent with Advanced Analytics Australia (Remote Contract) — 7-month overlap, split focus: Turquoise BNPL/risk/whisper vs Advanced edge CV*

Digikala

Tehran, Iran

AI Engineering R&D Lead (Concurrent)

Jan 2025 == Aug 2025

- Led team of 5 defining R&D roadmap + SLOs; shipped MCP with LangGraph cyclic graphs, zero-downtime hot-swapping, health-checked agent composition with deterministic fallback.
- Hybrid retrieval FAISS + Elasticsearch RRF + HyDE + KAG — +23% recall@10, -41% hallucination; multi-tier Redis caching, p99 <200ms.
- LLM serving on vLLM (PagedAttention) + Triton — 3.2x throughput, 60% cost reduction; 200 pods, MIG/HPA/VPA, Prometheus KV-cache dashboards.
- Partnered with infra/product establishing eval pipelines, A/B guardrails, canary rollouts.
- *Stack:* Python, LangGraph, LangChain, LlamaIndex, FAISS, Elasticsearch, ChromaDB, HyDE
- *Full-time lead; UBC/Trinity/Sharif were part-time remote (<10h/wk)*

HomaCloud

Tehran, Iran

MLOps Engineer (Founding) (Concurrent)

Aug 2021 == Feb 2022

- Provisioned multi-node K8s via Terraform; Ray + MLflow training pipelines, TF Serving + FastAPI — <10 min releases, 99.9% deploy success, 15+ models.
- GitOps with GitHub Actions, canary, spot-instance optimization; Prometheus/Grafana + ELK observability; registry + lineage.
- *Stack:* Kubernetes, Docker, Ray, MLflow, Terraform, ArgoCD, FastAPI, Prometheus
- *During B.Sc. — clean sequential after Mapna internship*

SELECTED PROJECTS & RESEARCH

Trinity College Dublin · EmoDub

May 2025 – Nov 2025

Emotion-aware voice translation

- SER front-end + wav2vec2.0 bullets (see experiences)

University of British Columbia

Feb 2024 – Nov 2025

Ultrasound report generation

EDUCATION

Sharif University of Technology

Tehran, Iran

M.Sc. Artificial Intelligence Computer Engineering (full-time, ongoing)

Sept 2025 – Present

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering GPA: 3.65

Sept 2020 – Sept 2025